

MAHMOUD AHMED EL-SHARAKY

Software Engineer | EdTech & Learning Platforms

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SUMMARY

Computer Science student (B.Sc., expected 2027) and self-taught software engineer who independently designs, builds, and deploys production systems used by real organizations. Full-stack across relational database design, REST API architecture, and React front-ends, grounded in data structures, algorithms, and software engineering. Shipped an enterprise HR platform serving **600+ employees across 25 branches** and an EdTech platform with an **8,000+ question bank**. Education technology is the focus: an 8,000+ question bank with adaptive mock exams, a weakness-analysis engine, and real-time learner analytics — designed, built, and operated solo, with measured learning outcomes rather than page views.

Target role: Software Engineer / Developer in university IT or EdTech — student and staff systems, e-learning and assessment platforms, and academic dashboards.

EXPERIENCE

Full-Stack Developer - SMLE Question Bank - EdTech Platform (Personal)

Feb 2025 - Present | Qassim, SA | [smle-question-bank.com](#)

- Built a platform for the Saudi Prometric (SCFHS) exam: an **8,000+ question bank**, adaptive mock tests, and a weakness-analysis engine with real-time analytics.
- Designed the PostgreSQL schema and REST API; users saw a **2.3x improvement** in measured proficiency over 3 months.
- Built a subscription/payments system on Moyasar (5-tier pricing, Apple Pay, webhook-driven fulfillment, VAT-inclusive invoicing) with a group-purchase flow that mints single-use invite links.
- Shipped a companion mobile app (Expo / React Native) sharing the same backend, alongside a Telegram bot for notifications and admin broadcast.

Full-Stack Software Engineer - Ultimate Care Rehabilitation Co. (Healthcare)

Oct 2025 - Present | Qassim, SA

- Architected and shipped a multi-branch HR platform (~20,000 LOC, 25 relational tables) covering employee lifecycle, payroll, attendance, documents, and transport for **600+ employees across 25 branches**.
- Implemented JWT authentication with role-based access control (RBAC), TTL-based in-memory caching, and composite database indexing - reaching **sub-second response times** and eliminating N+1 queries.
- Automated document-expiry alerts (30/60/90-day) and Arabic RTL PDF/Excel reporting with Hijri-Gregorian date conversion.

Web Developer (Freelance) - Erth Environmental Services

May 2025 - Jun 2025 | Qassim, SA | [erthfc.com](#)

- Built a bilingual (AR/EN) platform with automated consultation workflows and Google Maps integration; cut response time **40%** and lifted repeat requests **1.8x**.

Web Developer (Freelance) - Saleh Al-Hisouni Law Firm

Jul 2025 - Aug 2025 | Qassim, SA | [alhisony.com](#)

- Delivered a performant, SEO-optimized website with interactive 3D (Three.js), Schema.org structured data, and custom-domain email for formal correspondence.

EDUCATION

B.Sc. in Computer Science - Qassim University

2022 - 2027 (Expected) | Saudi Arabia

Relevant coursework: Data Structures, Analysis & Design of Algorithms, Operating Systems, Software Engineering, Databases, Logical Design, Coding & Encryption Theory.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), Python, SQL, HTML, CSS

Backend: Node.js, Express, REST API design, JWT / RBAC authentication, Nodemailer

Frontend: React 19, Vite, Framer Motion, Material-UI, Three.js / React Three Fiber

Databases: PostgreSQL (schema design, indexing, query optimization), Redis, Neon

Payments: Moyasar integration, webhook-driven fulfillment, subscription billing, VAT invoicing

Mobile: Expo, React Native

CS Foundations: Data Structures & Algorithms, Operating Systems, Software Engineering, Databases, Cryptography

Tools & Cloud: Git, GitHub, Vercel (serverless), AWS S3 / Cloudflare R2, Vercel Blob Storage, Axios, ESLint

LANGUAGES

Arabic (Native) • English (Fluent)